

Lab activity: Post-quantum cryptography (PQC) in practice

What you'll do

- Inspect real PQC artifacts and compare sizes.
- Extract performance numbers from a provided log.
- Explore a TLS capture showing a hybrid (classical + PQC) handshake in Wireshark.
- Draft a short migration plan using a checklist.

Provided files

The following files are included in the zip file attached to this lab.
You can also find the files inside `/home/coder/Desktop/exercise`.

- kyber_public.pem, kyber_private.pem
- dilithium_public.pem, dilithium_private.pem
- ciphertext.bin, signature.bin, message.txt
- perf_pqc_vs_classic.txt (captured performance log)
- tls_hybrid_demo.pcapng (Wireshark capture)
- Workbook: PQC-Lab-Workbook.xlsx.

Part A: Artifact exploration (point-and-click only)

Goal

Feel the practical size differences of PQC versus classical artifacts by looking at real files.

Activity

1. Open the Files panel and navigate to `pqc_assets/`
2. Right-click each file → Properties → note Size (bytes/KB):
 - kyber_public.pem, kyber_private.pem
 - dilithium_public.pem, dilithium_private.pem
 - ciphertext.bin, signature.bin
3. Enter your observations in the workbook 'Artifacts and Sizes' sheet.

Optional

Open the `.pem` files in the text editor and see that they're standard PEM containers (no edits needed).

Summary

You've practised recognising PQC key/ciphertext/signature footprint without touching code.

Part B: Performance (from a ready log, not running anything)

Goal

Compare recorded PQC performance versus RSA/ECC without benchmarking yourself.

Activity

1. Open perf_pqc_vs_classic.txt in the text editor.
2. Find the sections for: Kyber512 (KeyGen/Encaps/Decaps), Dilithium2 (Sign/Verify), RSA-2048 (Sign/Verify), ECDSA-P256 (Sign/Verify).
3. Copy the numbers into the workbook 'Performance Logs' sheet (one line per metric).
4. In the 'Notes' column, add quick takeaways (e.g., 'Verify faster/slower than...').

Summary

You've practised extracting real-world PQC perf signals with no terminal needed.

Part C: TLS hybrid handshake (read-only in Wireshark)

Goal

See what a real hybrid negotiation looks like on the wire.

Activity

1. Launch Wireshark from your lab environment.
2. Open tls_hybrid_demo.pcapng
3. In the Display filter, type: tls and press Enter.
4. Click the first ClientHello.
 - In the middle pane, expand TLS → Extensions → supported_groups / key_share
 - Note if you see a hybrid group (e.g., X25519+Kyber or similar naming from the capture).
5. Click the ServerHello.
 - Record the selected group (what the server actually chose).
6. Find the Certificate message (if present).
 - Inspect the Signature Algorithm field (e.g., a PQ signature like Dilithium, or a classical one if the demo uses hybrid key exchange only).
7. Enter each observation in the workbook 'TLS Trace' sheet (it tells you exactly where to look).

Summary

You've practised recognising PQC/hybrid parameters in a TLS handshake without configuring anything.

Part D: Migration mini-plan (checklist)

Goal

Sketch a practical transition plan to PQC.

Activity

1. Open the workbook 'Migration Plan' sheet.
2. For each row (Inventory, Risk, PQ Readiness, Hybrid, Performance, Interop, PKI, Rollout, Fallback):
 - add a brief Task owner, Due date and a one-line Note specific to your context (e.g., 'Pilot oqs-provider in staging L7 proxies').
3. Keep it succinct. This is a working checklist, not an essay.

Summary

You've practised mapping PQC concepts to concrete rollout steps.

Optional

1. Open message.txt, signature.bin, ciphertext.bin in the editor just to see how signatures/ciphertexts look and how large they are.
2. Skim headers inside the .pem files for format familiarity.

Wrap-up

- A filled workbook capturing PQC sizes, performance extracts, a TLS trace reading and a draft migration checklist.
- Practical familiarity with PQC artifacts and network traces, without writing a single line of code.